

LEMON-Factor: Bidirectional Factor-Level Semantic Alignment for Graph–Text Pairs

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Abstract. Graph–text evaluation requires more than checking whether the right entity names occur in a fluent sentence. A faithful verbalization must preserve the roles and relation meanings encoded by graph predicates, and a useful text-to-graph reconstruction must recover the same predicate-level information. We propose LEMON-FACTOR, a bidirectional factor-level diagnostic framework for graph–text pairs. LEMON-FACTOR decomposes relation predicates into weighted semantic factors and evaluates both KG→Text coverage and Text→KG recoverability. The framework is positioned between triple-level factuality metrics, which typically issue one decision per triple, and atomic-fact metrics, which decompose generated text rather than source predicates. On a stratified WebNLG pilot, expanded Forward LEMON-FACTOR reaches 0.7915 and Reverse LEMON-FACTOR reaches 0.7789. A MINE-STYLE adaptation inspired by KGGen’s Measure of Information in Nodes and Edges shows that node information is easier to recover than edge information: on a locked 50-fact subset, deterministic node information is 0.9600 and edge information is 0.5400. Controlled perturbation calibration shows that relation deletion reduces Forward LEMON-FACTOR by 0.211 while exact entity-label coverage changes by only 0.002. We further instantiate the same factor inventory format for DrugProt and BC5CDR and score controlled biomedical perturbations. The results support LEMON-FACTOR as an interpretable pilot framework for diagnosing relation-level semantic alignment. We treat rule-based factor inventories and synthetic LLM adjudication resources as provisional, not as human expert validation.

Keywords: Graph-to-text evaluation · Semantic alignment · Knowledge graphs · WebNLG · Biomedical relation extraction · Predicate decomposition

1 Introduction

Knowledge graphs encode information as typed relations between entities. In RDF-style graph-to-text generation, an input edge (s, p, o) is not only a pair of entity names. The predicate p states which relation holds, which semantic roles the arguments fill, and which part of the meaning must be preserved in text. The same issue appears in the reverse direction. If a text is converted back into

a graph, a useful reconstructed edge must preserve relation meaning, not merely mention the same entities.

This distinction is often hidden by holistic automatic metrics. A sentence may mention *Alan Bean* and *Wheeler, Texas* while omitting that the relation is *birthPlace*. In biomedical text, the same chemical and protein can be present while the polarity changes from inhibition to activation. Standard WebNLG evaluation uses reference metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT, together with human criteria such as data coverage, relevance, correctness, and fluency [1–3]. These criteria are useful, but they do not say which internal semantic component of a predicate is covered or missing.

We propose LEMON-FACTOR, a source-driven diagnostic framework for graph–text semantic alignment. The core idea is to decompose each source predicate into typed, weighted semantic factors. For example, *birthPlace* can be represented by a person-like subject factor, a biographical-relation factor, and a place-like object factor. A DrugProt inhibition predicate can be represented by a chemical actor, a gene/protein target, negative regulation, directionality, and evidence form. LEMON-FACTOR then checks whether these factors are expressed in text or recovered in a reconstructed graph. This creates a diagnostic layer between binary triple-level metrics, such as FactSpotter, and text-driven atomic-fact metrics, such as FActScore [10, 11].

The framework is bidirectional. In the forward direction, $\text{KG} \rightarrow \text{Text}$, it measures whether a text covers the semantic factors of the input graph. In the reverse direction, $\text{Text} \rightarrow \text{KG}$, it measures whether a reconstructed graph preserves the semantic factors of the source graph. We also implement a MINE-STYLE node/edge baseline inspired by KGGen’s Measure of Information in Nodes and Edges (MINE), which evaluates text-to-KG information retention through recovered nodes and relations [16]. The baseline is used as a transparent comparison point; it is not a reproduction of the full KGGen/MINE benchmark.

The paper asks five research questions. **RQ1:** Does factor-level coverage capture graph-to-text semantic preservation beyond entity-label overlap? **RQ2:** Does the same factor representation support both forward coverage and reverse recoverability? **RQ3:** How does factor-level recoverability relate to node/edge information retention in a MINE-style baseline? **RQ4:** Do the metrics behave predictably under meaning-destroying and meaning-preserving perturbations? **RQ5:** Does the same deterministic factor format produce useful perturbation diagnostics on biomedical graph–text data?

The contributions are: (1) a bidirectional, weighted factor-level formulation for graph–text semantic alignment; (2) a reproducible WebNLG pilot with factor decomposition resources, forward and reverse scores, and audit trails; (3) fixed factor inventories for WebNLG, DrugProt, and BC5CDR; (4) a MINE-style node/edge information baseline adapted to graph–text pairs; and (5) controlled perturbation scoring showing that relation-level damage is not captured by entity preservation alone. The study is deliberately framed as a diagnostic pilot. Factor decompositions are deterministic resources supported by rules and schema

constraints; synthetic LLM adjudication is treated as provisional development material, not as human-validated gold.

2 Related Work

2.1 RDF-to-text benchmarks and reference metrics

The WebNLG Challenge established RDF-to-text generation as a benchmark task in which systems map DBpedia-derived RDF triples to natural language [1]. The WebNLG+ shared task extended the setting to bilingual and bidirectional generation and included both automatic and human evaluation [2]. The GEM benchmark further argued for broader evaluation practice and challenge sets for NLG evaluation [3]. The standard WebNLG automatic stack includes reference-based metrics such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT [4–6]. These metrics produce useful holistic similarity signals, but they do not localize errors to a source predicate or to a semantic role inside the predicate.

2.2 Source-aware coverage and factuality metrics

A second line of work evaluates generated text against input data. PARENT weights n-gram precision and recall by evidence in the source table and reference text [7]. Data-QuestEval adapts question-answering evaluation to data-to-text semantic coverage without requiring a reference [8]. AlignScore learns a unified alignment function for factual consistency across several supervision sources [9]. The closest graph-to-text predecessor is FactSpotter, which decides whether a given triple (s, p, o) is present in a generated text and reports strong correlations with human judgments for data correctness, coverage, and relevance [10]. LEMON-FACTOR complements this line by moving from a binary whole-triple decision to a weighted decomposition of the predicate into semantic factors.

2.3 Atomic facts, semantic graphs, and directionality

Atomic factuality metrics decompose generated text into verifiable claims. FActScore evaluates long-form text by decomposing it into atomic facts and checking support against a knowledge source [11]; VERIScore extends claim verification for long-form factuality [12]. This direction is text-driven: the output text is decomposed into claims. LEMON-FACTOR is source-driven: the source predicate is decomposed into factors that can be checked in text or in a reconstructed graph.

Semantic graph metrics such as Smatch compare graph structures, especially AMR graphs, by maximizing overlap under node alignment [13]. S²match and Smatch++ add softer or more standardized graph comparison variants [14, 15]. These metrics are important for parser evaluation, but they presuppose graph representations on both sides. LEMON-FACTOR instead evaluates factor evidence from graph-text pairs and only uses reconstructed graphs in the reverse experiment.

Table 1. Positioning of LEMON-FACTOR relative to related evaluation families.

Family	Representative methods	Granularity	Missing diagnostic dimension	Direction
Surface/reference	BLEU, METEOR, chrF++, TER, BERTScore, BLEURT	sentence-level holistic	source-predicate roles	KG→Text
Source-aware	PARENT, QuestEval, Score	Data- n-gram, question, Align- sentence	explicit within-predicate factors	KG→Text
Triple-level	FactSpotter	binary triple	which part of the predicate failed	KG→Text
Atomic fact	FactScore, VERISCORE	text-derived atomic claims	source predicate decomposition	Text→KB
Node/edge	KGGen/MINE	node and edge retention	weighted predicate factors	Text→KG
LEMON-FACTOR	this work	weighted factors inside predicates	provisional resource, not human gold	bidirectional

2.4 Text-to-KG extraction and MINE

Recent text-to-KG work has renewed interest in evaluating whether extracted graphs retain useful information from plain text. KGGen introduces a language-model-based text-to-KG generator and releases MINE, Measure of Information in Nodes and Edges, as a benchmark for testing whether an extractor produces useful KGs from text [16]. Our MINE-STYLE baseline is inspired by this node/edge information split. It is not a reproduction of the KGGen benchmark, which targets KG extractors on plain text; rather, it adapts the idea to WebNLG graph-text pairs so that Text→KG recoverability can be compared with Reverse LEMON-FACTOR.

2.5 Behavioral testing and metric calibration

Evaluation metrics should be tested not only by correlation with human judgments, but also by whether they respond correctly to controlled changes. CheckList introduced behavioral tests for NLP models, including invariance tests, where label-preserving perturbations should not change predictions, and directional expectation tests, where perturbations should change predictions in a prescribed direction [17]. Perturbation CheckLists for NLG metrics apply the same idea to automatic metric evaluation [18]. He et al. use synthetic stress tests to expose blind spots of model-based NLG metrics [19]. Our calibration follows this methodology: relation deletion is a directional test, while entity aliasing and predicate paraphrase templates are invariance-style checks whose quality is explicitly marked.

3 Method

3.1 Problem setting

Let a graph-text pair be (G, T) , where G is a set of directed edges $e = (s, p, o)$ and T is a verbalization. The forward task asks whether the text covers the

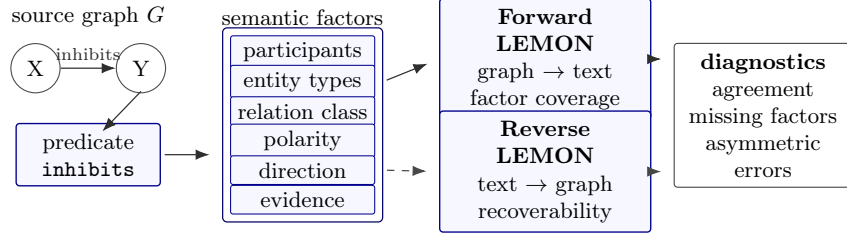


Fig. 1. Predicate-factor evaluation for graph-text alignment. A source predicate is decomposed into semantic factors, which are checked as forward graph-to-text coverage and reverse text-to-graph recoverability. Solid arrows denote forward evidence checks; dashed arrows denote reverse recovery checks.

information in the graph. The reverse task asks whether a graph reconstructed from the text, denoted $\hat{G}(T)$, preserves the information in the original graph. LEMON-FACTOR uses the same predicate-level semantic representation in both directions (Fig. 1).

3.2 Predicate factorization

Each predicate p is mapped to a weighted decomposition:

$$D(p) = \{(f_i, r_i, w_i)\}_{i=1}^n,$$

where f_i is a semantic factor, r_i is its role, and $w_i > 0$ is a salience weight. Roles include *predicate meaning*, *subject domain*, *object domain*, *value domain*, and modifier or background roles. The schema uses broad factors such as *person*, *place*, *organization*, *time*, *quantity*, and relation factors such as *biographical relation*, *location relation*, *part-whole relation*, *political relation*, and *creative relation*. This representation is intentionally coarser than a full ontology but richer than a single predicate label.

For example, *birthPlace* receives a person-like subject factor, a biographical predicate factor, and a place-like object factor. In the biomedical inventories, *chemical_inhibits_gene_or_protein* receives chemical and gene/protein role factors, an inhibition factor, negative polarity, directionality, and evidence-form factors. A text that mentions both entities but omits the relation cue should therefore receive partial credit, not full coverage. This is the key distinction from whole-triple binary scoring.

3.3 Forward LEMON: KG→Text coverage

For an edge e , text T , and component (f_i, r_i, w_i) , an evidence function $E(f_i, r_i, e, T) \in [0, 1]$ estimates whether the component is expressed. In the deterministic pilot, entity and domain factors are supported by normalized label or token evidence,

while relation factors are supported by predicate cues and factor cues. The edge-level forward score is:

$$S_F(e, T) = \frac{\sum_i w_i E(f_i, r_i, e, T)}{\sum_i w_i}.$$

Example-level and corpus-level scores are averages over edges with available decompositions. The implementation stores covered and missing factors for each edge, so the scalar score is accompanied by a diagnostic trace.

3.4 Reverse LEMON: Text→KG recoverability

The reverse direction first constructs a deterministic reconstructed graph $\hat{G}(T)$ from the text using entity-label evidence and predicate lexical cues. Reverse LEMON-FACTOR then compares each gold edge with its best recovered counterpart. Subject and object recovery contribute through node/domain evidence; predicate recovery contributes through overlap between the gold predicate’s factors and the recovered edge evidence. The reverse score therefore asks whether the source predicate’s semantic factors survive the text-to-graph reconstruction.

The current reconstructed graph is a lexical pseudo-extractor, not a full KG extraction model. This choice makes the pilot reproducible and keeps the focus on metric behavior rather than extractor quality. Future work can replace $\hat{G}(T)$ with KGGen, GraphRAG-style extraction, OpenIE, or domain-specific extractors.

3.5 MINE-style node/edge baseline

To compare factor-level recoverability with node/edge information retention, we implement a MINE-STYLE baseline inspired by KGGen’s MINE. Each gold triple is treated as a fact, and the reconstructed graph is searched for matching or related nodes and edges. The deterministic score separates node information, edge information, and a composite partial-credit node/edge score. We report it as a local baseline rather than a reproduction of the original benchmark.

3.6 Controlled perturbation calibration

A metric that measures semantic alignment should degrade under meaning-destroying perturbations and remain comparatively stable under meaning-preserving perturbations. We therefore create deterministic perturbations of both text and reconstructed graphs. Text-side destructive perturbations include relation phrase deletion, object swaps, and predicate swaps. Graph-side destructive perturbations include node drops, edge drops, and predicate swaps. Preserving perturbations include alias-like rewrites and template paraphrases, which are flagged as clean or weak-template depending on their reliability. All perturbed files and operation-level manifests are saved, enabling audit of each injected change.

Table 2. Fixed predicate-factor inventories. Coverage is computed against the current evaluation files.

Dataset	Factors	Predicates	Data predicates	Coverage
webnlg	24	121	119	1.000
drugprot	25	13	12	1.000
bc5cdr	25	1	1	1.000

4 Data and Experimental Resources

4.1 WebNLG pilot

The WebNLG experiments use a stratified English pilot built from RDF-to-text examples. The train split contains 200 examples and the development split contains 100 examples. Both splits cover 16 WebNLG categories. The development set contains 308 graph edges. The pilot is intentionally small: it validates the end-to-end scoring pipeline, exposes interpretable failure modes, and keeps all perturbation manifests auditable before scaling to larger graph-text resources.

4.2 Biomedical transfer resources

The biomedical transfer checks use two relation extraction corpora converted into the same graph-text JSONL format. DrugProt provides chemical-gene/protein relations with a rich label set, including inhibition, activation, agonist, antagonist, regulator, substrate, product, and part-of relations. In the current processed subset, the train split contains 500 abstracts, 3588 edges, 12 observed predicate types, and an average text length of 1729.5 characters. BC5CDR provides document-level chemical-induced disease relations. The converted train and development splits each contain 500 documents, with 985 and 961 CID edges respectively. These resources are used as transfer diagnostics only; we do not train or claim a new biomedical relation extraction system.

4.3 Predicate-factor inventories

The final experiments use fixed JSON predicate-factor inventories. Each inventory contains a controlled factor vocabulary and one weighted decomposition per predicate. WebNLG uses the fixed development inventory. DrugProt adds biomedical factors for chemical actor, gene/protein target, relation family, polarity, directionality, and evidence form. BC5CDR uses a compact document-level inventory for the chemical-induced disease relation, where the central factors are chemical exposure, disease outcome, causality, document-level evidence, and direction from chemical to disease. The validator checks that predicate coverage is complete for the current evaluation files and that weights sum to one for each predicate.

The inventories are deterministic resources. Earlier LLM-generated decomposition candidates are kept in the repository as development material, but they are not treated as expert gold and are not used as the main evidence in the deterministic scoring tables below.

5 Experiments

The experiments are organized by the five research questions from the introduction. All scoring runs are reproducible from JSONL inputs and fixed JSON factor inventories. No LLM calls are made in the deterministic perturbation scoring reported in RQ5.

5.1 RQ1: Forward factor-level semantic coverage

We first score WebNLG development texts using predicate decompositions and lexical cues. We then run an error-driven resource expansion: missing predicates are identified, rule-based or fallback decompositions are added, lexical cues are expanded, and coverage is recomputed. This measures how much of the initial low score comes from missing resources rather than from the text itself.

5.2 RQ2: Bidirectional factor recoverability

To test whether the factor representation is bidirectional, we reconstruct a graph from each WebNLG text with a deterministic lexical pseudo-extractor and compute Reverse LEMON-FACTOR. This score is compared with expanded Forward LEMON-FACTOR. A small forward–reverse gap would suggest that the same factor schema can support both $\text{KG} \rightarrow \text{Text}$ coverage and $\text{Text} \rightarrow \text{KG}$ recoverability.

5.3 RQ3: MINE-style node/edge information retention

We compare Reverse LEMON-FACTOR with a MINE-STYLE baseline inspired by KGGen’s Measure of Information in Nodes and Edges. The deterministic variant reports node information, edge information, and a composite node/edge score. A previously locked 50-fact subset is also used to show how binary fact recoverability differs from a partial-credit node/edge composite. These values are reported as diagnostic baselines, not as a reproduction of the KGGen/MINE benchmark.

5.4 RQ4: Controlled perturbation calibration

We inject deterministic noise into text and reconstructed graphs. Destructive perturbations should reduce relevant scores; preserving perturbations should produce smaller changes. Text-side perturbations are used primarily to calibrate Forward LEMON-FACTOR, while graph-side perturbations are used primarily to calibrate Reverse LEMON-FACTOR and MINE-style recoverability.

Table 3. Forward LEMON-FACTOR before and after decomposition and cue expansion.

Metric	Initial	Expanded	Delta
Scored edges	140	308	+168
Missing decomposition rate	0.5455	0.0000	-0.5455
Predicate cue coverage	0.3149	0.6526	+0.3377
LEMON-FACTOR coverage	0.3683	0.7915	+0.4232
Low-coverage edges	191	41	-150

5.5 RQ5: Cross-domain deterministic baseline scoring

The final perturbation files cover WebNLG, DrugProt, and BC5CDR. The operators keep the source graph fixed and change only the text, which lets us test whether a metric reacts to the intended semantic damage rather than to a change in the reference graph. The final scoring uses five operators: node deletion, relation-cue deletion, argument swap, polarity flip, and relation blur. Polarity flips are most informative for DrugProt and BC5CDR, where relation labels explicitly encode activation, inhibition, regulation, or causality.

We score the shared perturbation files with deterministic baselines. The goal is not to reproduce external systems such as FactSpotter or KGGen/MINE, but to provide local comparison points with the same inputs. We report entity recall, label match, a binary triple-style match, a MINE-STYLE node/edge harmonic score, a label-only LEMON variant, and the full deterministic LEMON-FACTOR proxy based on the fixed factor inventories and perturbation metadata. The same scoring layer is used for the compact ablation study.

6 Results

6.1 RQ1: Forward coverage and resource completeness

The initial factor dictionary covered only part of the WebNLG development predicates. As a result, the first coverage run scored 140 of 308 edges and produced a missing decomposition rate of 0.5455. Resource expansion reduced the missing rate to zero and increased LEMON-FACTOR coverage from 0.3683 to 0.7915 (Table 3). This increase is not presented as evidence that all texts are faithful. It shows that factor-level evaluation depends on explicit predicate decompositions and lexical evidence resources.

The comparison with lexical controls (Table 4) clarifies the signal. Exact entity-label overlap is already high and unchanged by expansion. Graph-text token cosine and Jaccard are also unchanged. LEMON-FACTOR changes because it uses predicate decompositions and role-aware evidence. This supports the interpretation that LEMON-FACTOR measures a different diagnostic layer than surface overlap.

Table 4. Baseline comparison on the WebNLG development pilot.

Method	Initial	Expanded	Delta
Exact entity-label overlap	0.9215	0.9215	0.0000
Predicate lexical cue coverage	0.3149	0.6526	+0.3377
Graph-text token cosine	0.7357	0.7357	0.0000
Graph-text token Jaccard	0.6233	0.6233	0.0000
LEMON-FACTOR coverage	0.3683	0.7915	+0.4232

Table 5. Bidirectional comparison on the WebNLG development pilot.

Method	Direction	Factors	Retrieval	Score	Diagnostic level
Forward LEMON-FACTOR	KG→Text	yes	no	0.7915	factor/edge/example
Reverse LEMON-FACTOR	Text→KG	yes	no	0.7789	factor/edge/example
MINE-style composite node/edge	Text→KG	no	yes	0.8028	node/edge/fact
MINE-style LLM fact recoverability	Text→KG	no	yes	0.5200	fact/subgraph
Graph-text token cosine	KG↔Text	no	no	0.7357	corpus
Graph-text token Jaccard	KG↔Text	no	no	0.6233	corpus

6.2 RQ2: Bidirectionality

The reverse experiment recovered 390 of 408 nodes and 196 of 308 edges. Reverse LEMON-FACTOR reached 0.7789, close to the expanded Forward LEMON-FACTOR score of 0.7915 (Table 5). The small gap suggests that the predicate-factor representation can be used in both directions: from graph to text and from text back to graph. The result should be interpreted as a pilot observation because the reconstructed graph is produced by a lexical pseudo-extractor.

6.3 RQ3: Relation information is harder than node information

The deterministic MINE-STYLE composite score is 0.8028, but this aggregate hides a strong node/edge gap. On the locked 50-fact subset, node information is 0.9600, while edge information and deterministic fact recoverability are both 0.5400; the partial-credit composite is 0.7500. A separate LLM fact check gives a close score of 0.5200 and agrees with the deterministic fact decision on 0.9800 of the cases. These values indicate that WebNLG texts tend to preserve entity mentions more reliably than relation recoverability, and that partial-credit node/edge scores and binary fact scores should not be treated as interchangeable.

6.4 RQ4: Perturbation calibration

Table 6 reports targeted calibration summaries after separating text-side and graph-side perturbations. Destructive text perturbations produce a monotonic Forward LEMON-FACTOR decrease, while destructive graph perturbations produce monotonic decreases in Reverse LEMON-FACTOR and MINE-style recoverability. This supports the expected directional behavior of the metrics.

The clearest sanity check is relation phrase deletion (Table 7). Exact entity-label coverage changes by only -0.0020, while Forward LEMON-FACTOR drops

Table 6. Targeted perturbation calibration. Slopes are average score changes with increasing destructive noise.

Metric	Target	Destr. slope	Spearman	Viol.	Pres. delta
Forward LEMON-FACTOR	text	-0.3241	-1.0000	0	0.0527
Exact label coverage	text	-0.1508	-0.9162	1	0.0585
Predicate cue coverage	text	-0.4525	-0.6667	0	0.0438
Reverse LEMON-FACTOR	graph	-0.5546	-1.0000	0	0.0000
MINE composite node/edge	graph	-0.5161	-1.0000	0	0.0000
MINE fact recoverability	graph	-0.7204	-1.0000	0	0.0000

Table 7. Relation deletion sanity check.

Metric	Baseline	Final	Delta
Exact label coverage	0.9215	0.9194	-0.0020
Forward LEMON-FACTOR	0.7915	0.5808	-0.2106
Predicate cue coverage	0.6526	0.2890	-0.3636
MINE fact recoverability	0.6364	0.2825	-0.3539

by -0.2106. This is the central validity signal of the pilot: entity preservation alone is not relation preservation, and the factor-level metric reacts to relation-level semantic omission.

6.5 RQ5: Cross-domain deterministic scoring

Table 8 reports compact deterministic scores on the shared perturbation files for WebNLG, DrugProt, and BC5CDR. The full WebNLG and biomedical tables are generated as separate repository artifacts. Entity recall remains relatively high for relation blur and polarity changes, especially in BC5CDR, because the same biomedical entities are still present. Triple-style and MINE-style scores are lower when the relation cue or edge evidence is damaged, but they do not identify which semantic component was affected.

The deterministic LEMON-FACTOR proxy is most informative when the factor inventory contains the tested semantic dimension. This is why WebNLG polarity flip is not penalized strongly: many WebNLG predicates in the current pilot do not encode polarity. In DrugProt and BC5CDR, polarity and direction are part of the task semantics, so argument swap and polarity flip reduce LEMON-FACTOR full more clearly. The result supports the main diagnostic claim: factor inventories make the tested meaning dimensions explicit, while entity preservation alone can hide relation-level damage.

The compact ablation in Table 9 confirms that the factor groups are not interchangeable. Removing participant-role factors produces the largest sensitivity loss, especially on WebNLG and DrugProt, where argument structure is central to relation meaning. Removing polarity or evidence factors also reduces sensitivity in the biomedical setting. Label-only scoring can be harsh when surface cues disappear, but it cannot identify whether the damage concerns roles, direction, polarity, or evidence.

Table 8. Compact deterministic scoring summary over controlled perturbations.

Dataset	Perturbation	Entity	Triple	LEMON-full
WebNLG	Node deletion	0.5142	0.0857	0.6603
WebNLG	Relation cue deletion	0.7287	0.1643	0.7031
WebNLG	Argument swap	0.7212	0.2977	0.7080
WebNLG	Polarity flip	0.7237	0.3010	1.0000
WebNLG	Relation blur	0.7237	0.1643	0.7031
DrugProt	Node deletion	0.5281	0.1459	0.7000
DrugProt	Relation cue deletion	0.5867	0.1604	0.7543
DrugProt	Argument swap	0.5869	0.1869	0.6019
DrugProt	Polarity flip	0.5872	0.1774	0.5285
DrugProt	Relation blur	0.5867	0.1604	0.7543
BC5CDR	Node deletion	0.8321	0.5288	0.6000
BC5CDR	Relation cue deletion	0.9659	0.5412	0.6924
BC5CDR	Argument swap	0.9696	0.7280	0.5500
BC5CDR	Polarity flip	0.9667	0.7222	0.5512
BC5CDR	Relation blur	0.9659	0.7208	0.6924

Table 9. Factor-group ablation: mean controlled-damage drop ($1 - score$); higher is more sensitive.

Ablation	BC5CDR	DrugProt	WebNLG
LEMON-full	0.3828	0.3322	0.2451
Label-only	0.2716	0.4445	0.5352
Unweighted	0.3508	0.3267	0.2537
- roles	0.2228	0.2125	0.1188
- direction	0.3728	0.3123	0.2451
- polarity	0.3583	0.3074	0.2451
- evidence	0.3357	0.2903	0.2451

6.6 RQ6: LLM reliability probe

As an exploratory reliability check, we ran three 45-item judge batches (135 judgments) with `google/gemini-2.0-flash-001`, `openai/gpt-4o-mini`, and `meta-llama/llama-3.1-70b-instruct`. Each judge saw the source edge, factor inventory, and perturbed text, and returned JSON-schema constrained labels: covered, partial, or absent.

Table 10. Exploratory LLM factor-judging probe. Scores: covered=1, partial=0.5, absent=0.

Subset	Mean LLM	Judge agr.	Det. agr.
All judged factors	0.809	0.597	0.561
Argument swap	0.765	0.386	0.432
Node deletion	0.731	0.502	0.408
Polarity flip	0.857	0.766	0.694
Relation blur	0.907	0.730	0.647

Table 10 shows that LLM judges often recover factors from residual context. The mean factor score is high (0.809), while agreement with deterministic perturbation labels is moderate (0.561). Agreement is lowest for argument swap and

node deletion, so we treat LLM judging as a diagnostic complement rather than human validation.

7 Error Analysis

The experiments reveal two recurring error classes. The first is resource incompleteness: predicates without decompositions or with weak lexical cues cannot be scored reliably. The error-driven expansion step was designed to expose and reduce this problem. After expansion, all 308 development edges had decompositions, but fallback decompositions remain less informative than expert-designed ones.

The second class is relation recoverability. Node information is high because WebNLG verbalizations often repeat entity labels, but edge information is substantially lower. This pattern appears in the reverse experiment, in the MINE-style node/edge gap, and in the relation-deletion perturbation. It suggests that the difficult part of graph-text alignment is not usually naming entities, but preserving and detecting the relation that links them.

The factor-level trace is useful here. A low LEMON-FACTOR can be traced to a predicate, a factor role, a missing cue, or a weak subject/object evidence signal. This is the main practical advantage of the framework: it turns a scalar score into a debugging signal for datasets, generators, and extraction pipelines.

8 Limitations

The main limitation is the absence of human expert validation. The factor inventories are deterministic resources based on predicate labels, schema constraints, rules, and development-time LLM candidates. The three-judge LLM probe checks residual recoverability, but it is not human gold. Prior work reports LLM-judge biases, so future work should add expert annotation and inter-annotator agreement [23–25].

A second limitation is lexical evidence. The deterministic scorer can find labels and known cues, but it can miss implicit relations, long-range entailment, and paraphrases outside the cue dictionary. Future versions should add embedding, entailment, or QA-based evidence while preserving the factor-level audit trail.

The reverse pipeline is also limited. The reconstructed graph is produced by a lexical pseudo-extractor, not by a full text-to-KG system. Likewise, our MINE-style baseline adapts KGen’s node/edge information idea to graph-text pairs; it is not a reproduction of the original MINE benchmark.

Finally, the perturbations are deterministic and auditable, but still template-like. They support initial construct-validity checks, especially relation deletion, argument swap, and polarity flip, but they do not prove robustness to natural paraphrase. DrugProt and BC5CDR are used as transfer diagnostics for roles, directionality, and polarity; we do not claim state-of-the-art biomedical relation extraction performance.

9 Conclusion

We introduced LEMON-FACTOR, a bidirectional factor-level diagnostic framework for graph–text pairs. The method decomposes relation predicates into weighted semantic factors and evaluates both $\text{KG} \rightarrow \text{Text}$ coverage and $\text{Text} \rightarrow \text{KG}$ recoverability. In a WebNLG pilot, expanded Forward LEMON-FACTOR reached 0.7915 and Reverse LEMON-FACTOR reached 0.7789, which supports the use of one predicate-factor representation in both directions.

The MINE-style comparison showed that node information is easier to recover than edge information. Controlled perturbation calibration showed that relation deletion strongly reduces factor-aware scores while leaving exact entity-label coverage nearly unchanged. The same factor inventory format was then instantiated for DrugProt and BC5CDR, where polarity, directionality, and causality are explicit relation dimensions. These findings position LEMON-FACTOR as a diagnostic layer between triple-level factuality and atomic-fact evaluation. Future work will replace synthetic adjudication with expert annotation, add stronger paraphrase evidence, and extend the biomedical transfer checks to full human-validated factor coverage studies.

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